Extra Bite #13: Dynamic Range and Intermodulation

Blocking dynamic range, intermod, desensitization, third-order intercept, preselectors, and link-budget math.

Group E4D · 13 pool questions · Do after: EC-B · 5-10 minute presentation · Practice exams at eham.net/exams

Dynamic Range Vocabulary

A real receiver has to find weak signals while strong signals are sitting nearby. **Dynamic range** is how wide a window of input levels the receiver can handle cleanly. Too small a dynamic range, and strong adjacent signals create spurious products that mask the weak one you actually want.

Term	Definition	Q
Blocking dynamic range	**The difference in dB between the noise floor and the level of an incoming signal that will cause 1 dB of gain compression** (E4D01). The point where one strong nearby signal starts squashing the receiver's gain.	E4D01
Problems from poor dynamic range	**Spurious signals caused by cross-modulation and desensitization from strong adjacent signals** (E4D02).	E4D02
Desensitization	**Reduction in receiver sensitivity caused by a strong signal near the received frequency** (E4D06). The strong signal compresses the front-end gain; weak signals get quieter.	E4D06
How to reduce desensitization	**Insert attenuation before the first RF stage** (E4D07). Brings the strong signal back into the linear region.	E4D07

Intermodulation

What causes intermodulation in an electronic circuit? Nonlinear circuits or devices (E4D08). Any nonlinearity (a saturating amp, a hot mixer) mixes two input signals and produces sum-and-difference products at new frequencies.

Question	Answer	Q
Cause of intermod between two nearby repeaters	**Output signals mix in the final amplifier of one or both transmitters** (E4D03). A second carrier sneaks back into one transmitter's final and mixes with its own carrier.	E4D03
Fix for nearby-transmitter intermod into a repeater	**A properly terminated circulator at the output of the repeater's transmitter** (E4D04). A circulator passes the wanted signal out but absorbs anything coming back in.	E4D04
Why do **odd-order** intermod products matter most?	**Odd-order products of two in-band signals are also likely to be in the band** (E4D11). $2f_1 - f_2$ lands close to f_1, so it gets through the receive filter.	E4D11

Third-Order Intercept (IP3)

A receiver's IP3 (third-order intercept point) is a single-number figure of merit for its handling of strong signals.

What does a third-order intercept level of 40 dBm mean? A pair of 40 dBm input signals will theoretically generate a third-order intermodulation product that has the same output amplitude as either of the input signals (E4D10). It is an extrapolated point that real signals never reach; higher IP3 = the receiver tolerates more before intermod becomes a problem.

Preselectors

What is the purpose of the preselector in a communications receiver? To increase the rejection of signals outside the band being received (E4D09). A bandpass filter ahead of the mixer that keeps out-of-band giants from reaching the active stages.

Link Budget Math

A link budget tracks gain and loss in dB from transmitter to receiver. Gains add, losses subtract. The leftover is the received signal level. **Link margin** = received level minus the threshold (MDS + required SNR).

E4D13 (received level): TX +40 dBm + TX ant 6 dBi + RX ant 3 dBi − path 100 dB = **−51 dBm**.
E4D12 (link margin): received = 40 + 10 − 3 − 136 = −89 dBm; threshold = −103 + 6 = −97 dBm; margin = **+8 dB**.
E4D14 (dBm → watts): 0 dBm = 1 mW; −100 dBm = 10^{-10} mW = **0.1 picowatt**.

Mental Cheat Sheet

- Blocking dynamic range = noise floor → 1 dB gain compression.

Common Traps

- Blocking DR is **1 dB compression**, not the third-order intercept.

- Poor dynamic range → cross-mod + desensitization from strong adjacent signals.
- Desensitization = strong nearby signal reduces RX sensitivity.
- Fix for desensitization: **attenuator before the first RF stage**.
- Intermod comes from **nonlinearities**.
- Repeater intermod fix: **circulator on the TX output**.
- Odd-order intermod products land **in** the band → why they matter.
- IP3 = extrapolated point where input = intermod output.
- Preselector = bandpass filter ahead of the mixer.
- dBm: 0 dBm = 1 mW; −100 dBm = 0.1 pW.
- Link margin = received level − required level (MDS + SNR).

- Desensitization is fixed by **attenuating**, not by raising IF or front-end gain.
- Intermod comes from **nonlinearity**, not feedback (positive or negative).
- Repeater intermod fix is a **circulator**, not a bandpass filter alone.
- Preselector job is **out-of-band rejection**, not "selecting the RF amp device" or "storing frequencies."
- −100 dBm is **0.1 picowatt**, not microwatts. Watch the unit prefix carefully.

Bite 13 Question Bank (E4D)

E4D01. What is meant by the blocking dynamic range of a receiver?
A. The difference in dB between the noise floor and the level of an incoming signal that will cause 1 dB of gain compression.
B. The minimum difference in dB between the levels of two FM signals that will cause one signal to block the other.
C. The difference in dB between the noise floor and the third-order intercept point.
D. The minimum difference in dB between two signals which produce third-order intermodulation products greater than the noise floor.

E4D02. Which of the following describes problems caused by poor dynamic range in a receiver?
A. Spurious signals caused by cross modulation and desensitization from strong adjacent signals.
B. Oscillator instability requiring frequent retuning and loss of ability to recover the opposite sideband.
C. Poor weak signal reception caused by insufficient local oscillator injection.
D. Oscillator instability and severe audio distortion of all but the strongest received signals.

E4D03. What creates intermodulation interference between two repeaters in close proximity?
A. The output signals cause feedback in the final amplifier of one or both transmitters.
B. The output signals mix in the final amplifier of one or both transmitters.
C. The input frequencies are harmonically related.
D. The output frequencies are harmonically related.

E4D04. Which of the following is used to reduce or eliminate intermodulation interference in a repeater caused by a nearby transmitter?
A. A band-pass filter in the feed line between the transmitter and receiver.
B. A properly terminated circulator at the output of the repeater's transmitter.
C. Utilizing a Class C final amplifier.
D. Utilizing a Class D final amplifier.

E4D06. What is the term for the reduction in receiver sensitivity caused by a strong signal near the received frequency?
A. Reciprocal mixing.
B. Quieting.
C. Desensitization.
D. Cross modulation interference.

E4D07. Which of the following reduces the likelihood of receiver desensitization?
A. Insert attenuation before the first RF stage.
B. Raise the receiver's IF frequency.
C. Increase the receiver's front-end gain.
D. Switch from fast AGC to slow AGC.

E4D08. What causes intermodulation in an electronic circuit?
A. Negative feedback.
B. Lack of neutralization.
C. Nonlinear circuits or devices.
D. Positive feedback.

E4D09. What is the purpose of the preselector in a communications receiver?
A. To store frequencies that are often used.
B. To provide broadband attenuation before the first RF stage to prevent intermodulation.
C. To increase the rejection of signals outside the band being received.
D. To allow selection of the optimum RF amplifier device.

E4D10. What does a third-order intercept level of 40 dBm mean with respect to receiver performance?
A. Signals less than 40 dBm will not generate audible third-order intermodulation products.
B. The receiver can tolerate signals up to 40 dB above the noise floor without producing third-order intermodulation products.
C. A pair of 40 dBm input signals will theoretically generate a third-order intermodulation product that has the same output amplitude as either of the input signals.
D. A pair of 1 mW input signals will produce a third-order intermodulation product that is 40 dB stronger than the input signal.

E4D11. Why are odd-order intermodulation products, created within a receiver, of particular interest compared to other products?
A. Odd-order products of two signals in the band being received are also likely to be within the band.
B. Odd-order products are more likely to overload the IF filters.
C. Odd-order products are an indication of poor image rejection.
D. Odd-order intermodulation produces three products for every input signal within the band of interest.

E4D12. What is the link margin in a system with a transmit power level of 10 W (+40 dBm), a system antenna gain of 10 dBi, a cable loss of 3 dB, a path loss of 136 dB, a receiver minimum discernable signal of −103 dBm, and a required signal-to-noise ratio of 6 dB?
A. −8 dB.
B. −14 dB.
C. +8 dB.
D. +14 dB.

E4D13. What is the received signal level with a transmit power of 10 W (+40 dBm), a transmit antenna gain of 6 dBi, a receive antenna gain of 3 dBi, and a path loss of 100 dB?
A. −51 dBm.
B. −54 dBm.
C. −57 dBm.
D. −60 dBm.

E4D14. What power level does a receiver minimum discernible signal of −100 dBm represent?
A. 100 microwatts.
B. 0.1 microwatt.
C. 0.001 microwatts.
D. 0.1 picowatts.

Answer Key

E4D01: A	E4D02: A	E4D03: B	E4D04: B
E4D06: C	E4D07: A	E4D08: C	E4D09: C
E4D10: C	E4D11: A	E4D12: C	E4D13: A
E4D14: D			